1. Do the vectors $\begin{bmatrix} 1 \\ 1 \\ 1 \\ t \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \\ -1 \\ t \end{bmatrix}$ form an angle of $\pi/3$ in $\mathbf{R}^4$ for any values of t? If so, find the value of t.

2. Consider three planes in $\mathbf{R}^3$ space that go through the origin and are respectively perpendicular to $\begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$. Determine whether these planes intersect in just the origin, an entire line, or a complete plane.

3. Find a mapping whose image is the hyperplane of solutions to the equation

$$x_1 + 2x_2 - 3x_3 + 4x_4 = 5$$

in $\mathbf{R}^4$. Also, find out whether the mapping is linear or not.

4. Given a triangle T which contains $\begin{bmatrix}1\\1\\1\end{bmatrix}$, $\begin{bmatrix}1\\2\\3\end{bmatrix}$, and $\begin{bmatrix}3\\5\\1\end{bmatrix}$.

4.1. Find an equation of the plane on $\mathbf{R}^3$ that contains a triangle T.

4.2. Find an equation of a line (ray) on $\mathbf{R}^3$ that **never intersect** the triangle T.

4.3. Find the intersection point between a line $l = \left\{ \begin{bmatrix}3\\-3\\0\end{bmatrix} + \begin{bmatrix}-1\\9\\2\end{bmatrix} s \;\middle|\; s \in \mathbf{R} \right\}$ and the plane from 4.1.

5. From the question 4, we saw that 3 points on $\mathbf{R}^3$ forming a triangle will always lying on some plane on $\mathbf{R}^3$. Can we say that 4 points on $\mathbf{R}^3$ will always form a rectangle that lies on some plane on $\mathbf{R}^3$?

6. Row problem vs. column problem:

   6.1. The equations $-x + 2y = 3$ and $x + 2y = 1$ both describe lines in the plane (using the standard Euclidean coordinate system). Sketch both lines and find their intersection.

   6.2. What linear combination of $(-1, 1)$ and $(2, 2)$ adds up to $(3, 1)$ in $\mathbf{R}^2$? Make a sketch to illustrate your answer.